

VAIBHAV KANDPAL

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SUMMARY

Computer Science undergraduate (2024-28) with experience in full-stack development and machine learning. Proficient in building **scalable web applications** (React.js, FastAPI, Node.js) and **end-to-end ML pipelines** for real-time data systems. Delivered AI-driven, production-ready solutions achieving **98% model accuracy**; ranked **Top 5 in India** at NASA Space Apps Challenge 2024 (823 teams).

EDUCATION

B.Tech in Computer Science & Engineering

Dr. Akhilesh Das Gupta Institute of Professional Studies, Delhi

[2024-2028]

CERTIFICATIONS & COURSES

Data Science & AI - IIT Roorkee

6-month professional programme

Ongoing

Data Analytics Job Simulation - Deloitte Australia

Forage certified programme

July 18, 2025

TECHNICAL SKILLS

Programming Languages: Python, JavaScript, C, C++

Frontend: React.js, HTML, CSS, Tailwind CSS

Backend: FastAPI, Node.js, Django, Express.js

Databases: MySQL, SQLite, TimescaleDB

ML / Data: Scikit-learn, NumPy, Pandas, Random Forest, Explainable ML

Tools: Git, GitHub, REST APIs, WebSockets, Redis, Celery

PROJECTS & EXPERIENCE

ORBITAIR - AI-Based Air Quality Forecasting System

Group project - solely responsible for data scraping, backend, frontend & database; ML model training by teammates.

- Built a **scalable FastAPI backend** with TimescaleDB for geospatial & time-series data ingestion and analytics
- Developed a **data fusion pipeline** integrating satellite (TEMPO), sensor (EPA, OpenAQ) and NOAA datasets
- Created a **React + Leaflet dashboard** for real-time pollution visualization
- Achieved **98% prediction accuracy** using explainable ML models; optimized for real-time ingestion at scale

ORBIT-OPS - ML-Based Satellite Visibility Prediction

Group project - solely responsible for data scraping, backend, frontend & database; ML model training by teammates.

- Designed an **ML pipeline** using orbital (TLE) data; generated **48K+ samples** via a self-supervised data pipeline
- Built a **<10ms latency inference system** supporting real-time satellite visibility decisions
- Enabled automated **handover optimization** & ground station selection

ACHIEVEMENTS

- Top 5 in India - NASA Space Apps Challenge 2024** (823 competing teams)